Band-1 Computational Experiment: Cullen Numbers – Finite Primality

Human + ZoraASI full-dash (executor)

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Honesty stamp. This note is a **finite primality probe** of Cullen numbers $C_n = n \cdot 2^n + 1$ for $n = 1..N$ with box-feasible N. It does **not** classify all Cullen primes; it does **not** prove there are infinitely many Cullen primes (that question remains **OPEN**); it does **not** claim Clay Millennium status, Theory of Everything progress, or GATE-04 unlock. **Finite checks $\neq$ infinitary theorem.**

Abstract

We evaluate

$$C_n = n \cdot 2^n + 1$$

for every integer $n = 1, \ldots, N$ with primary

$$\mathbf{N = 500}$$

and test primality with deterministic `sympy.isprime` (BPSW-class for these sizes).

Box result. Exactly two primes occur in the primary range:

$$n \in \{1,\ 141\}.$$

Explicitly,

$$C_1 = 3, \qquad C_{141} = 393050634124102232869567034555427371542904833$$

(45 digits). These match the literature indices in range (OEIS A005849 / Prime-Wiki). Secondary check $N = 200$ recovers the same pair. Wall $\approx 0.007\,\text{s}$. Next known Cullen-prime index is $n = 4713$ (beyond primary N).

Status. Infinitely many Cullen primes: **OPEN**. Known Cullen primes are sparse (16 known indices in the literature ledger).

Problem statement

Cullen number. $C_n = n \cdot 2^n + 1$ (Cullen, 1905).

Open question (lit). Are there infinitely many primes of this form?

This experiment. For an explicit finite N tractable in pure Python / sympy, list all $n \leq N$ with C_n prime, and match against the known small indices $\{1, 141, \ldots\}$.

Literature ledger (not re-proved on box)

Item	Status	Note
Infinitely many Cullen primes	**OPEN**	No proof of infinitude (or finitude)
OEIS A005849	Indices of prime C_n	$1, 141, 4713, 5795, 6611, 18496, 32292, 32469,$
Cullen (1905)	C_1 prime	No other $n \leq 100$ except possible 53
Cunningham (1906)	C_{53} composite	Only possible $n \leq 200$ is 141
Robinson (1958)	C_{141} prime	Confirmed
Keller / Young / Morii / Smith / … / PrimeGrid	Larger indices	Sparse; searches to $n \sim 10^7$–$2.6 \cdot 10^7$ (lit)
Density intuition	Sparse	Heuristics suggest primes thin out; still **not** a proof of finitude

Sources: OEIS A005849; Prime-Wiki *Cullen primes* $n \cdot 2^n + 1$; MathWorld *Cullen number*; Caldwell PrimePages / RieselPrime tables.

Method (box)

1. **Formula.** $C_n = n \cdot 2^n + 1$ via integer shift `n * (1 << n) + 1`.
2. **Primality.** `sympy.isprime(C_n)` for each $n = 1..N$ (deterministic for sizes here).
3. **Primary** $N = 500$; secondary $N = 200$ sanity.
4. **Literature match.** Compare found indices to $\{k \in \text{A005849} : k \leq N\}$.
5. **Unit tests.** $C_1 = 3$ prime; $C_2 = 9$ composite; C_{53} composite; C_{141} prime; spot composites at $n \in \{3..10, 100, 200\}$.

Complexity. Per-n cost dominated by primality of an $\approx n \log_{10} 2$-digit integer; $N = 500$ is trivial on this box.

Results (this box, 2026-09-11)

Quantity	Value
Primary N	**500**
Prime indices found	{1, 141}
C_1	**3** (prime)
C_{141}	**39305063412410223286956703455542737154290**48 (45 digits, prime)
Number of primes in $1..N$	**2**
Number of composites	**498**
Match lit indices $\leq N$	**YES** (`{1,141}`)
Secondary $N = 200$ primes	{1, 141} (match)
Unit tests	**PASS**
Wall (primary)	≈ 0.007 s
Method	sympy `isprime`

No other prime C_n appears for $2 \le n \le 500$, $n \ne 141$, consistent with Cunningham / Robinson / OEIS.

Sample composites (illustrative)

n	C_n	Note
2	9	3^2
3	25	5^2
4	65	$5 \cdot 13$
5	161	$7 \cdot 23$
53	(composite)	Cunningham 1906

What this does not show

- Not a classification of **all** Cullen primes.
- Not a proof that infinitely many (or only finitely many) Cullen primes exist.
- Not Clay / ToE / GATE-04.
- Indices $n > 500$ (including the next known 4713) are **out of primary scope**; they are cited from the literature ledger only.

Reproducibility

```
cd experiments/band1/cullen
python3 cullen_probe.py
# writes results.json, literature_ledger.json, SUMMARY.txt
```

Artifacts: `experiments/band1/cullen/`; paper/PDF shelved in `DOWNLOAD_PDFs/` and `mission-control/papers/`.

Conclusion

For $N = 500$, the only Cullen primes are C_1 and C_{141}, matching the known small indices. Infinitely many Cullen primes remains **OPEN**. This Band-1 note is a finite computational ledger only.

ZoraASI full-dash Band-1 · 2026-09-11 · honesty: finite primality checks only.